

**Before the United States Copyright Office
Library of Congress
Washington, D.C. 20024**

**iFixit Comment Regarding
Software-Enabled Consumer Products**

Docket No. 2015-6

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Interests of iFixit

iFixit is the international, open-source, online repair manual for everything. The organization is devoted to making “repair accessible and easy for as many people around the world as possible.”¹ Repair saves consumers money and protects the environment. As a trade, repair also represents more than 3 million domestic jobs.

iFixit represents a global community of makers, hardware hackers, fixers, tinkerers, and repair professionals. The iFixit community is over one million people strong—and the collaborative group has published nearly 20,000 repair guides on iFixit.com. In 2015, this massive, free resource has helped tens of millions of people around the globe fix everything from mobile phones to game consoles, toys to musical instruments. As a founding member of The Repair Association, iFixit also stands firm in its support of tinkerers and independent repair professionals.

Introduction

The gradual and now ubiquitous integration of software into consumer products has challenged the ability of the current copyright regime to address increasingly complex issues of ownership, access, and authorization.² Every product—from primitive to cutting-edge—eventually breaks. But most products can also be repaired.

¹ *Why Repair?* iFixit <https://www.ifixit.com/Info/why> (last visited Feb. 9, 2016).

² See WHITE PAPER ON REMIXES, FIRST SALE, AND STATUTORY DAMAGES 50 (Jan. 2016) *available at* <http://www.uspto.gov/sites/default/files/documents/copyrightwhitepaper.pdf>. [hereinafter WHITE PAPER].

Cars are an important example. When a car's maintenance light flashes on its dashboard, traditionally, the owner could open the hood and search for the problem. Owners could make repairs for themselves, if they were sophisticated enough—perhaps consulting a manual to guide the process or identify replacement parts. Failing that, the owner could also shop around for a mechanic (either with a dealership or from an independent repair shop), receive quotes and estimates, and pay a mechanic to repair what is broken. Copyright law has stymied this traditional model of self-reliance and market competition for repairs—in ways Congress and the Constitution³ did not intend.

It's been years since cars were mere mechanical contrivances. A modern car is equipped with dozens of computers running hundreds of thousands of lines of computer code. Because software and firmware are copyrightable, manufacturers can assert unprecedented levels of control over the aftermarket and repair of vehicles. If *only* dealerships have the ability and the tools necessary to access, modify, and understand that code, then only manufacturers will be able to make repairs necessary for complex, modern vehicles. A well-publicized incident involved a farmer, Kerry Adams, who bought a set of tractors that contained copyright-protected software and oppressive contract provisions that blocked Kerry from making repairs himself, instead having to fly in a factory technician to make repairs as authorized by the manufacturer.⁴ Because the equipment-

³ CONST. art. I, § 8, cl. 8 (promoting the “Progress of Science and useful Arts”).

⁴ Kyle Wiens, *New High-Tech Farm Equipment is a Nightmare for Farmers*, WIRED (March 18, 2013), <http://www.wired.com/2013/03/you-dont-own-your-cellphones-or-your-cars/>; see Laura Sydel, *DIY Tractor Repair Runs Afoul of Copyright Law*, NATIONAL PUBLIC RADIO (Aug. 25, 2015), <http://www.npr.org/sections/alltechconsidered/2015/08/17/432601480/diy-tractor-repair-runs-afoul-of-copyright-law>; Kyle Wiens, *Forget the Cellphone Fight—We Should be Allowed to Unlock Everything We Own*, WIRED (Mar. 18, 2013), <http://www.wired.com/2013/03/you-dont-own-your-cellphones-or-your-cars/>.

maker restricted access to the copyrighted software, the farmer could not use a local mechanic. The equipment and its maintenance became so cost prohibitive, Kerry abandoned the equipment altogether.

This problem is not limited to vehicles; all software-enabled products suffer from analogous issues, and this has a profound and negative effect on consumers and repair professions.

Recently, a catastrophic software error in Apple's iPhones has inspired the ire of millions of Apple users, because it is systematically limiting users' ability to repair their iPhones. Error 53 affects late model iPhones and other Touch ID-enabled devices.

Most often, Error 53 affects users who have repaired their home buttons (or its flex cable), or had them repaired by a third party. The phone works fine after a successful repair, sometimes for months. As soon as users connect to a computer and attempt to update the phone's operating system, Apple's Error 53 effectively kills their previously functioning phone. Only Apple-authorized repair providers have access to the tools that can re-validate a home button with a display screen—tools that they will not share with independent repair professionals.⁵ Because of copyright law, independent repair professionals cannot create their own tools to repair the devices.

⁵ Kyle Wiens, *What's Up With Error 53?*, iFIXIT (Feb. 5, 2016), <https://ifixit.org/blog/7889/whats-up-with-error-53/>.

Indeed, there are countless examples⁶ of the strange interactions between the law's agenda for protecting software copyrights and unintended consequences for hardware and consumers. This all raises the question: Should copyright law permit this?

The ability for the public to repair its products promotes creativity and reduces ecological and economic waste. Although repair has historically been reasonably permissible or shielded by legal principles such as fair use⁷ or the first sale doctrine,⁸ the introduction of software has frustrated this. iFixit proposes that copyright law expand its exemptions to include repairing software-enabled devices, thus returning the right to repair back to the consumer as well as independent repair shops.

The Ability to Repair Benefits Consumers, Markets, and the Environment

There is remarkable value in repair—running the gamut from the personal to ecological. Tinkering promotes self-reliance and creativity. Do-It-Yourself culture (“DIY”), tinkering, or “maker” culture has steadily grown in popularity, thanks to the ability of makers to find one

⁶ These include various trades from the agricultural to the medical as well as primary and secondary (reseller) markets. *See generally, We Are Repair*, THE REPAIR ASSOCIATION <http://repair.org/industries/main/> (last visited Feb. 9, 2016).

⁷ *See e.g., Lewis Galoob Toys, Inc. v. Nintendo of America, Inc.*, 964 F.2d 965 (9th Cir. 1992); *Sega Enters. Ltd. v. Accolade, Inc.*, 977 F.2d 1510 (9th Cir. 1992). The Digital Millennium Copyright Act's (“DMCA”) rulemaking procedure was also intended as an analogous safety valve, like fair use, to ensure copyright law does not over burden the public interest. *See also Exemptions to Prohibition Against Circumvention*, 37 C.F.R. § 201.40(b)(6) (2015).

⁸ 17 U.S.C. § 109. *But see, Rental Amendment of 1984*, Pub. L. No. 98-450, 98 Stat. 1727 (Oct. 4, 1984), *amending* 17 U.S.C. § 109; *Capital Records, LLC v. ReDIGI Inc.*, 934 F. Supp. 2d 640 (S.D.N.Y. 2013). WHITE PAPER at 38 (discussing various stakeholders' interpretation of the purpose of the first sale doctrine); *id.* at 62 n.380.

another online and provide advice and assistance that was never available to the inventors of a previous age. Tinkering promotes deep engagement with technology, learning, and exploration—culminating in a well-deserved sense of pride and satisfaction with one’s work. Contests, competitions, and trade shows are hosted around the world to promote the repair ethos.

Some of the best and most promising fixers are young people. Adrian Mayberry fixed his police department’s bomb disposal robot,⁹ and 15-year-old Kelvin Doe from Sierra Leone has built everything from a broadcast station to generators out of repurposed parts.¹⁰ This entrepreneurial spirit carries on through their lives, incubating the innovative industries of the future.

Repair itself can extend the life of a product so that consumers can use it even after it breaks. Critically, repairing products is far more economical than replacement. Independent repair shops and DIY repairs are usually more cost-effective than repairs made within “authorized” or dealer repair networks. Easing restrictions on repair opens up competition at the secondary market by reducing barriers to entry for independent repair shops, which are currently hampered by overly-burdensome contract provisions and intellectual property laws.

Finally, the freedom to repair devices produces several environmental benefits. When a product breaks and is discarded, it often ends up in a landfill. There, it either remains inert as garbage or emits pollutants that run off into our ecosystem. This problem is getting worse. The

⁹ Julia Bluff, *Tinkering Teen Repairs SWAT Robot for Police*, iFixIT (Jan. 25, 2016), <http://ifixit.org/blog/7809/tinkering-teen/>.

¹⁰ Hayley Hudson, *Kelvin Doe, Self-Taught Engineering Whiz From Sierra Leone, Wows MIT Experts*, HUFFINGTON POST (Nov. 19, 2012), http://www.huffingtonpost.com/2012/11/19/kelvin-doe-self-taught-en_n_2159735.html.

International Environmental of the United Nations Environmental Program Technology Center projects that e-waste increases 40% each year.¹¹ Repair helps diminish this trend; it is a natural ally of the old mantra “reduce, reuse and recycle.” It promotes sustainable practices that reduce waste.

The Current Copyright Regime Unreasonably Restricts The Public’s Ability to Repair Software-Enabled Devices

iFixit is primarily concerned with the public’s ability to access and repair the hardware they have bought—hardware they believe they own. Software is always installed on, and as a result inextricably a part of, the hardware that contains it. When it comes to embedded software, the hardware is useless without the software. Repairs to latter affect the functionality of the former. That is to say that many questions regarding the laws and regulations of software also regard its corresponding hardware.

As digital intellectual property issues began to boil over into the physical world, Congress passed the Digital Millennium Copyright Act (“DMCA”),¹² to address rightsholders’ concerns. The radical overhaul of the domestic copyright system accomplished its goals of providing greater protection to rightsholders in the digital era—but this came at the expense of shifting copyrights carefully weighed balance in favor of rightsholders at the expense of consumers.¹³ Critically, the DMCA’s safeguards were put in place to prevent the copying of entertainment media, such as

¹¹ Jim Lynch, *Snapshot of Worldwide Electronics Recycling 2013*, iFIXIT (May 7, 2013), <http://ifixit.org/blog/4662/snapshot-of-worldwide-electronics-recycling-2013/>.

¹² Pub. L. No. 105-304, 112 Stat. 2860 (1998).

¹³ *Murphy v. Millennium Radio Grp. LLC*, 650 F.3d 295, 393 (3rd Cir. 2011) (“[I]t is undisputed that the DMCA was intended to expand—in some cases . . . significantly—the rights of copyright owners.”).

music and movies. Instead, in the last twenty years, they have been widely utilized by manufacturers to keep control over their goods long past the point of sale. In terms of the repair market, manufacturers have wielded the DMCA to establish a repair monopoly over physical products with essential software.

An increasingly commonplace trend in copyright and anti-circumvention threats involves manufacturers bootstrapping the rights afforded to software to impose restrictions on the use of its (generally non-copyrightable) hardware.¹⁴ Courts have broadly accepted licenses and contract provisions that further limit consumer's ability to access hardware.¹⁵

Section 1201(c)3 of the DMCA attempts to limit the statute's scope solely to protecting rightholders' interests in software and attempts to avoid interfering with hardware.¹⁶ However, this section largely expresses Congress's intent to not proscribe, prescribe, or privilege any specific technological protection measure or circumventing device. This provision does not go far enough to preventing software's overflow into hardware.

¹⁴ See e.g., *Lexmark Int'l, Inc. v. Static Control Components, Inc.*, 387 F.3d 522 (6th Cir. 2004) (reuse of ink cartridges); *Chamberlain Grp., Inc. v. Skylink Techs., Inc.*, 381 F.3d 1178 (Fed. Cir. 2004) (universal garage door openers); *Storage Tech. Corp. v. Custom Hardware Eng'g & Consulting, Inc.*, 421 F.3d 1307 (Fed. Cir. 2005) (tape cartridge silos).

¹⁵ See *MDY Indus., LLC v. Blizzard Entertainment, Inc.*, 629 F.3d 928 (9th Cir. 2010).

¹⁶ See 17 U.S.C. § 1201(c)(3) ("Nothing in this section shall require that the design of, or design and selection of parts and components for, a consumer electronics, telecommunications, or computing product provide for a response to any particular technological measure, so long as such part or component, or the product in which such part or component is integrated, does not otherwise fall within the prohibitions of subsection (a)(2) or (b)(1).").

The Notice of Inquiry also illuminates how Congress took great care to ensure software repair was balanced in the new laws, but there is a conspicuous absence of discussion on the effects on hardware.

Congress revisited the issues surrounding software and copyright law with the DMCA. As particularly relevant here, the DMCA amended section 117 of the Copyright Act to permit the reproduction of computer programs for the purposes of machine maintenance or repair following a court of appeals decision that cast doubt on the ability of independent service organizations to repair computer hardware. This provision foreshadows the more general concerns raised by the spread of software in everyday products—namely, that maintaining or repairing a software-enabled product often will require copying of the software.¹⁷

Since software is a protected work under copyright law, Congress and the courts have been careful to carve out certain protections for downstream users of software.¹⁸ After *Mai Industries*, practitioners and commentators feared that the Ninth Circuit’s onerous interpretation of software infringement would limit computer repair. This so concerned Congress, it enacted section 117(c) in the DMCA to lift the restriction on software use. The DMCA’s amendments, although addressing issues at the software level, failed to effectively address similar repair issues the spilled over into the hardware.

Copyright Law Should Be Amended to Safeguard the Public’s Ability to Repair Its Software-Enabled Devices

Repair benefits everyone. Unfortunately, the collateral damage of copyright law’s overreach in the digital world stymies the public’s ability make repairs. The few carve-outs in the law are

¹⁷ *Software-Enabled Consumer Products Study: Notice and Request for Public Comment*, 80 Fed. Reg. 77,668 at 77669 (Dec. 15, 2015) (citations omitted).

¹⁸ *See e.g.*, 17 U.S.C. §§ 106, 117, 1201(f).

simply too narrow and do not effectively safeguard hardware repair for software-enabled devices. Copyright law should include rigorous provisions expressly to ensure individuals and independent repair shops can access and modify hardware. Such exemptions would forward the public's interest while balancing and respecting the interests of rightsholders.

Conclusion

For the foregoing reasons, iFixit respectfully requests that the Agency and Legislature remain mindful of copyright law's effects on hardware repair, and take steps to promote the public's ability to repair its lawfully purchased devices.

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